AIRLINE TICKETING

FLIGHT CONNECTION GRAPH CRUD

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PROJECT TITLE

Airline Ticket Reservation System Using CRUD Operations

THIS PROJECT IS A SIMPLE AIRLINE TICKET RESERVATION SYSTEM THAT ALLOWS USERS TO MANAGE PASSENGER TICKET RECORDS USING CRUD OPERATIONS. THE SYSTEM HELPS STORE,RETRIEVE, UPDATE, AND DELETE TICKET INFORMATION EFFICIENTLY.

PROBLEM STATEMENT

MANAGING AIRLINE TICKET RESERVATIONS MANUALLY CAN BE DIFFICULT AND TIME-CONSUMING. LARGE AMOUNTS OF PASSENGER INFORMATION MUST BE STORED, UPDATED, AND ACCESSED REGULARLY. WITHOUT AN ORGANIZED SYSTEM,ERRORS SUCH AS DUPLICATE RECORDS, MISSING DATA, AND INCORRECT TICKET DETAILS MAY OCCUR.
THIS PROJECT AIMS TO CREATE A SIMPLE DIGITAL SYSTEM THAT MANAGES AIRLINE TICKET DATA EFFICIENTLY.

Real-world Application

Airline ticket reservation systems are widely used in the travel and aviation industries.
Applications include:

- Airline booking systems
- Airport reservation counters
- Travel agency management systems
- Online ticket booking platforms

These systems help airlines maintain passenger records, manage bookings, and improve customer service.

DATA STRUCTURE USED

THIS PROJECT USES A LINKED LIST DATA STRUCTURE TO STORE PASSENGER TICKET INFORMATION. A LINKED LIST ALLOWS DYNAMIC MEMORY ALLOCATION, MAKING IT SUITABLE FOR STORING RECORDS THAT CAN FREQUENTLY CHANGE. EACH NODE IN THE LINKED LIST REPRESENTS A PASSENGER TICKET AND CONTAINS DETAILS SUCH AS PASSENGER NAME, TICKET NUMBER, FLIGHT NUMBER, DESTINATION, AND SEAT NUMBER.

• SYSTEM DESIGN

The system follows a menu-driven design where users select options to perform different operations. Main modules include:

- Add Ticket
- View Ticket Details
- Update Ticket Information
- Delete Ticket Record
- Exit the Program

The user interacts with the system through a menu interface, and the system processes the selected operation.

The algorithm for the airline ticket reservation system works as follows:

ALGORITHM

Start the program.

Display the main menu with different options.

Accept the user’s choice.

Perform the selected operation (Add, View, Update, or Delete ticket).

Display the result of the operation.

Repeat the process until the user selects the exit option.

End the program

IMPLEMENTATION

The project is implemented using a programming language such as C/C++.
The program uses functions to perform CRUD operations:

- addTicket() to add new passenger details
- displayTickets() to show all ticket records
- updateTicket() to modify ticket information
- deleteTicket() to remove a ticket record

The system uses a linked list to store and manage all ticket data.

FEATURES OF THE SYSTEM

User-friendly Interface: The system provides a simple menu-driven interface that allows users to easily interact with the program.

- CRUD Operations: Supports Create, Read, Update, and Delete operations for managing passenger ticket records.
- Efficient Data Management: Stores and organizes passenger information in a structured way.
- Quick Access to Records: Users can easily search and view ticket details.
- Dynamic Data Handling: Uses a linked list data structure which allows flexible data storage and modification.

DEMO OUTPUT

```
PS C:\Users\padarthi sahithi\Desktop\DSA_project> gcc al.c
PS C:\Users\padarthi sahithi\Desktop\DSA_project> ./a.exe
Enter number of airports: 3
Enter airport 0 name: chennai
Enter airport 1 name: delhi
Enter airport 2 name: mumbai

------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 1
Enter source airport index: 0
Enter destination airport index: 1
Enter ticket price: 5000
Flight added successfully

------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 1
Enter source airport index: 0
Enter destination airport index: 2
Enter ticket price: 6500
Flight added successfully
```

```
------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 5

Flights from chennai:
 -> mumbai (Price: 6500) -> delhi (Price: 5000)

Flights from delhi:


Flights from mumbai:
```

```
------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 4
Enter source airport index: 0
Enter destination airport index: 1
Flight found! Ticket Price: 5000
```

```
------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 3
Enter source airport index: 0
Enter destination airport index: 1
Enter new ticket price: 5500
Flight price updated successfully

------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 5

Flights from chennai:
 -> mumbai (Price: 6500) -> delhi (Price: 5500)

Flights from delhi:


Flights from mumbai:
```

```
------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 2
Enter source airport index: 0
Enter destination airport index: 2
Flight deleted successfully

------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 5

Flights from chennai:
 -> delhi (Price: 5500)

Flights from delhi:


Flights from mumbai:
```

```
------ Airline Ticketing System ------
1. Add Flight
2. Delete Flight
3. Update Flight Price
4. Search Flight
5. Display Flights
6. Exit
Enter choice: 6
Data saved successfully. Exiting program...
PS C:\Users\padarthi sahithi\Desktop\DSA_project>
```

Advantages of the System

• Reduces Manual Work: Automates the process of managing airline ticket records.

• Improves Accuracy: Minimizes human errors in storing and updating passenger details.

• Efficient Record Management: Makes it easy to add, update, or delete ticket information.

• Time Saving: Faster access and processing of passenger ticket data.

• Scalable System: Can be expanded with additional features like database integration and online booking.

CONCLUSION

The Airline Ticket Reservation System provides a simple way to manage passenger ticket information using CRUD operations. It demonstrates the practical use of data structures in solving real-world problems.
The system helps in organizing ticket data efficiently and reduces errors in record management.

FUTURE WORK

The project can be further improved by adding additional features such as:

- Graphical User Interface (GUI)
- Online booking functionality
- Database integration (MySQL)
- Payment gateway integration
- Passenger login and authentication system

These improvements would make the system more advanced and closer to real airline reservation systems.